

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-SP-008

Title: Dust Control and Respiratory Protection in Sinter Plant

Department: Sinter Plant

Revision: 01

Date: 25-Apr-2024

1. PURPOSE

To control exposure of personnel to respirable dust and to ensure proper use of respiratory protection in the Sinter Plant.

2. EXPOSURE LIMITS

- Respirable dust in the work area must be below 3 mg/m³ (8-hour average)
- Areas above this limit are marked "Respirator Zone" with yellow signboards

3. ENGINEERING CONTROLS

- Electrostatic precipitator (ESP) must be in service whenever the sinter machine is running
- ESP outlet emission limit: 50 mg/Nm³; if exceeded for 30 minutes, inform the Environment Cell (Ext. 2650)
- Dedusting fans at transfer points must be checked every shift
- Water sprinklers on raw material conveyors to be kept ON in dry weather

4. RESPIRATORY PROTECTION

- Minimum FFP2 disposable respirator in Respirator Zones; FFP3 or half-mask with P3 filter for cleaning of ESP hoppers
- Fit check before each use; beards prevent a proper seal
- Replace disposable respirators every shift or when breathing becomes difficult

5. HOUSEKEEPING

- Use vacuum cleaning or wet cleaning only; dry sweeping with brooms and compressed-air blowing are prohibited

6. HEALTH SURVEILLANCE

All Sinter Plant personnel undergo lung function testing (spirometry) once a year at the Occupational Health Centre.